
MEASUREMENT NOTE

Insurance Costs and Household Financial Resilience

A Descriptive Measurement Note on the Hidden Fragility Premium

Arne Kotte

AUTHOR

Arne Kotte

DOCUMENT TYPE

Measurement note

DATE

May 2026

A Descriptive Measurement Note on the Hidden Fragility Premium

Author: Arne Kotte **Date:** May 2026 **Document type:** Descriptive public-data measurement note — revised after second review

Abstract

This paper reframes insurance as a household-resilience burden rather than a miscellaneous budget category. The central object is not an insurance price index alone, but recurring insurance and medical-risk costs scaled by income and, conceptually, liquid resources. Public aggregate data motivate the issue: from 1998 to 2025, motor vehicle insurance CPI rose 251.2 percent and hospital services CPI rose 296.7 percent, versus 97.5 percent for headline CPI. The revised contribution is distributional. Consumer Expenditure evidence for calendar-year 2024 shows selected reference-period insurance and medical-risk costs absorbing 4.8 percent of annual income for consumer units below \$25,000, compared with 1.4 percent above \$100,000. These estimates are point estimates and should be read as a descriptive burden screen, not as a fully annualized household-cost measure. MEPS evidence shows medical burden and access barriers concentrated among lower-income, uninsured, and high-burden groups. The results are descriptive, not causal, and do not validate the full liquidity-based fragility-premium object. They support a narrower conclusion: insurance and medical-risk exposure should be measured directly in household-resilience work and then tested against liquidity and hardship outcomes in future research.

I. Introduction

A household can be insured and still financially fragile. It can also face lower headline inflation while the cost of staying protected continues to rise. Vehicle insurance, health insurance, medical out-of-pocket costs, deductibles, and household coverage are not ordinary discretionary purchases. For many households they protect work continuity, transportation, housing stability, and access to care. When those costs rise, the household may remain formally protected while losing liquid resilience.

The earlier version of this paper treated that intuition mainly through aggregate price and spending trends. That was useful as motivation but too thin as an economic paper. The relevant object is not simply whether selected insurance categories rose faster than headline CPI. It is how recurring defensive costs absorb household resources, who bears that burden, and how burden interacts with liquidity and access barriers.

This measurement note therefore defines an **insurance-resilience burden**. In its feasible public-data form, the object is:

$$B_{it} = (\text{recurring insurance spending}_{it} + \text{medical out-of-pocket spending}_{it}) / \text{income}_{it}.$$

The conceptual target is broader:

$$FP_{it} = (\text{recurring premiums}_{it} + \text{expected out-of-pocket exposure}_{it} + \text{deductible/shock capacity}_{it}) / \text{liquid resources}_{it}.$$

The first object can be approximated only partially with Consumer Expenditure data. The second is the stronger household-finance object and requires additional evidence from MEPS, SHED, SCF, or administrative/state insurance data. This paper does not claim to estimate the full FP_{it} . It uses public evidence to show why that object is worth measuring.

The note makes four limited contributions. First, it separates insurance price growth, household spending, medical exposure, and liquidity burden. Second, it uses aggregate public data only as background evidence. Third, it adds distributional Consumer Expenditure evidence showing that selected insurance and medical-risk costs absorb a larger income share for lower-income consumer units. Fourth, it uses Medical Expenditure Panel Survey evidence to show why realized spending alone is incomplete: low out-of-pocket spending can coexist with delayed or unaffordable care, and higher burden groups report more access barriers.

The note remains descriptive. It does not identify the causal effect of insurance repricing on savings, borrowing, delinquency, or coverage choices. It does not estimate a monetary-policy channel. It does not claim that all insurance categories have risen faster than headline CPI. Its claim is narrower: household resilience analysis should treat recurring insurance and medical-risk costs as a measurable candidate fragility premium, and public data show that the income-scaled burden is distributional.

II. Conceptual framework: from insurance prices to resilience burden

An insurance price index is not a household resilience burden. A premium bill is closer, but still incomplete. A serious household object must include at least four elements.

First, there is the recurring premium or payment required to maintain coverage. This includes vehicle insurance, health insurance, renters or homeowners coverage, and other household protection costs where relevant. Second, there is residual out-of-pocket exposure: deductibles, coinsurance, uncovered services, claim denials, repair gaps, and medical bills not fully covered by insurance. Third, there is exposure or need. Vehicle insurance matters more for households whose work requires driving. Medical costs matter more for households with chronic conditions or poor

health. Property coverage matters differently by geography and tenure. Fourth, there is the denominator: income, cash flow after fixed obligations, liquid savings, or other resources available to absorb shocks.

The household-level object is therefore not simply:

insurance price growth > headline CPI.

It is closer to:

resilience burden = defensive recurring costs + residual shock exposure,
scaled by household capacity.

This distinction changes the empirical standard. Headline CPI is a useful reference point, but it cannot be the main denominator. A category can rise faster than CPI while becoming less burdensome if income and liquid resources rise faster. A category can rise slower than CPI while becoming more burdensome for households with concentrated exposure or stagnant income. The central question is distributional capacity, not price movement alone.

III. Data and measurement boundaries

The paper combines three public-data modules.

The first module uses BLS/FRED aggregate series as motivation. Monthly CPI series are converted to calendar-year annual averages. Consumer Expenditure all-consumer-unit spending series are annual nominal dollar measures. CPI rows are price indexes; CE rows are spending measures and should not be interpreted as pure prices. Because annual averages are used, seasonal adjustment differences are less consequential than in monthly comparisons, but they are still a measurement caveat. The tenants' and household insurance CPI series is not seasonally adjusted, while several other CPI series are seasonally adjusted.

The second module uses previously built Consumer Expenditure public-use work for calendar-year 2024. The unit is a **consumer unit**, not necessarily a household. The estimates group consumer units by annual income and measure selected reference-period insurance and medical-risk costs: health insurance, healthcare out-of-pocket spending, vehicle insurance, and housing insurance. Childcare is excluded from the core insurance-resilience measure because it is better understood as a broader work-continuity cost. These CEX estimates are point estimates without survey-design standard errors in the current note. They should be read as a conservative burden screen, not as a complete annual household-cost estimate.

The third module uses MEPS public-use evidence from prior medical-fragility work. MEPS is used to connect medical out-of-pocket burden, poverty/income group, insurance status, health status, and delayed or unaffordable care barriers. This module is descriptive. It helps validate the claim that realized spending alone does not capture vulnerability because some people report low spending precisely when care is delayed or unaffordable.

Several boundaries follow. The paper does not use the BLS health-insurance CPI as a main premium measure because that index uses an indirect retained-earnings method and is not a household premium bill. CE health-insurance spending is therefore more appropriate for the household-budget claim, while still not being a pure price index. The paper also narrows broad property-insurance claims: the national tenants' and household insurance CPI does not show the same long-run acceleration as auto insurance or hospital services, so geographic property-risk claims require better state-level data.

IV. Aggregate motivation: selected defensive costs have outpaced headline prices

Aggregate evidence motivates the question but does not answer it. From 1998 to 2025, headline CPI rose 97.5 percent. Motor vehicle insurance CPI rose 251.2 percent, medical care CPI rose 139.6 percent, and hospital services CPI rose 296.7 percent. Tenants' and household insurance CPI rose 66.1 percent, less than headline CPI, which is an important caution against a uniform insurance story.

Consumer Expenditure spending measures show another budget-facing pattern. From 1998 to 2024, vehicle-insurance spending rose 169.7 percent, health-insurance spending rose 344.1 percent, and healthcare spending rose 225.6 percent, compared with 92.4 percent headline CPI growth over the same window. These are spending changes, not price changes. They combine prices, utilization, coverage, household composition, and reporting changes.

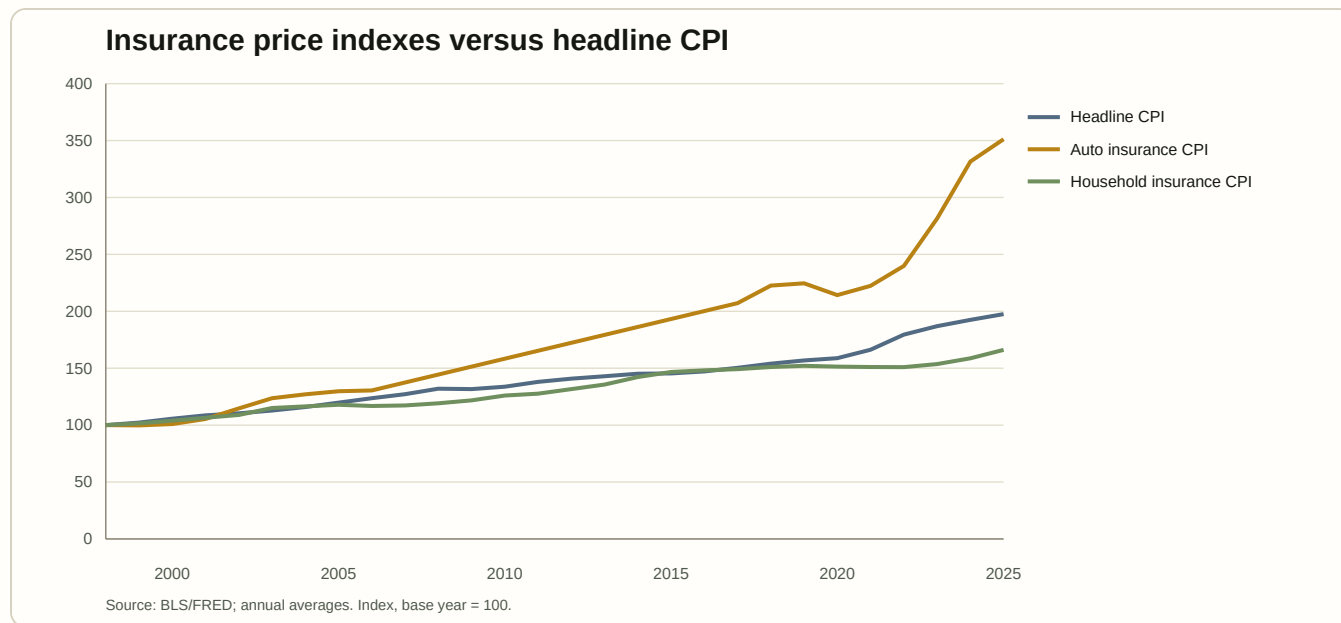


Figure 1. Insurance price indexes versus headline CPI.

Table 1. Growth in selected insurance and medical resilience-cost proxies. CPI rows are price indexes. Consumer Expenditure rows are nominal spending measures for all consumer units and are not price indexes.

Series	ID	Start	End	Series growth %	Series CAGR %	Headline CPI growth %	Headline CAGR %
Headline CPI	CPIAUCSL	1998	2025	97.5	2.55	97.5	2.55
Motor vehicle insurance CPI	CUUR0000SETE	1998	2025	251.2	4.76	97.5	2.55
Tenants' and household insurance CPI	CUUR0000SEHD	1998	2025	66.1	1.90	97.5	2.55
Medical care CPI	CPIMEDSL	1998	2025	139.6	3.29	97.5	2.55
Hospital services CPI	CUSR0000SEMD	1998	2025	296.7	5.24	97.5	2.55
Vehicle insurance spending	CXU500110LB0101M	1998	2024	169.7	3.89	92.4	2.55
Health insurance spending	CXUHLTHINSRLB0101M	1998	2024	344.1	5.90	92.4	2.55
Healthcare spending	CXUHEALTHLB0101M	1998	2024	225.6	4.65	92.4	2.55

The table is best read as a source of motivation. It shows that some defensive categories have moved differently from broad consumer prices, but it does not by itself establish household burden. The remainder of the paper therefore shifts from price growth to burden shares.

V. Consumer Expenditure evidence: burden is larger at the bottom

The Consumer Expenditure burden exercise moves closer to the relevant object, but it remains a point-estimate measurement screen. In calendar-year 2024 public-use estimates, selected reference-period insurance and medical-risk costs averaged \$782 for consumer units below \$25,000 of income, \$1,266 for \$25,000–\$50,000, \$1,564 for \$50,000–\$100,000, and \$2,357 for \$100,000 or more. Dollar costs rise with income, but income-scaled burden falls. The selected-cost burden equals 4.8 percent of annual income below \$25,000, 3.5 percent for \$25,000–\$50,000, 2.2 percent for \$50,000–\$100,000, and 1.4 percent above \$100,000.

These percentages should be interpreted carefully. The numerator is a selected CEX reference-period cost screen, not a fully annualized premium ledger. The denominator is annual income. For that reason, the table is best read as conservative descriptive evidence that lower-income consumer units face a higher income-scaled burden from selected protection costs, not as a final annual burden estimate.

Table 2. CEX 2024 selected insurance/medical resilience-cost burden by consumer-unit income group. Point estimates use selected 2024 reference-period Consumer Expenditure public-use records and annual income groups. The numerator excludes childcare and combines health insurance, healthcare out-of-pocket, vehicle insurance, and housing insurance. Because this is a reference-period burden screen rather than a fully annualized household-cost estimate, the figures should be read as descriptive lower-bound evidence, not as annual premium burden.

Income group	N CUs	Mean selected cost	Cost / income	Cost / total expenditure
< \$25k	2,952	\$782	4.8%	19.2%
\$25k-\$50k	3,643	\$1,266	3.5%	21.3%
\$50k-\$100k	4,844	\$1,564	2.2%	18.7%
>= \$100k	7,090	\$2,357	1.4%	16.2%
Nonpositive / missing	45	\$1,375		23.4%

This pattern is the core distributional fact in the revised note. Insurance and medical-risk costs are not merely higher or lower in dollar terms; they absorb different shares of household capacity. A lower-income consumer unit may spend less in dollars but still lose a larger share of income to maintaining basic protection.

The component table clarifies the burden mix. Health insurance, healthcare out-of-pocket spending, vehicle insurance, and housing insurance all rise in dollar terms with income. Childcare is intentionally excluded from the core measure. It may matter for work continuity, but including it inside an insurance-resilience burden makes the concept too broad.

Table 3. CEX 2024 selected insurance/medical resilience-cost components by consumer-unit income group. Dollar entries are conservative mean selected costs by component. Childcare is excluded from the core measure and reserved for possible broader continuity-cost work.

Income group	Health insurance	Healthcare OOP	Vehicle insurance	Housing insurance
< \$25k	\$407	\$132	\$175	\$68
\$25k-\$50k	\$584	\$233	\$322	\$128
\$50k-\$100k	\$730	\$277	\$410	\$146
>= \$100k	\$1,059	\$516	\$560	\$222
Nonpositive / missing	\$703	\$343	\$224	\$104

The CEX evidence is still not the full fragility premium. It does not directly measure liquid assets, deductible exposure, claim risk, coverage adequacy, geographic repricing, or outcomes such as emergency borrowing and missed payments. It also does not separate all consumer units from exposed or insured consumer units in the way a final paper should. The positive-cost selection

issue is substantive: zero observed spending may reflect no exposure, no coverage, delayed payment, underinsurance, or reporting/reference-period timing. A future version should report both all-unit burden and exposed-unit burden.

VI. Medical-risk evidence: realized spending misses access barriers

Medical risk is part of the insurance-resilience burden because insurance status and out-of-pocket exposure shape whether health shocks become financial shocks. MEPS evidence adds an important warning: realized spending can understate fragility when people delay or forgo care.

In 2022 MEPS person-level estimates, the OOP medical burden and access barriers are both concentrated among lower-income groups. Poor or negative-income persons had a mean OOP/income burden of 3.40 percent, an OOP ≥ 10 percent income share of 10.9 percent, and an any-access-barrier rate of 20.4 percent. High-income persons had a mean burden of 0.68 percent, an OOP ≥ 10 percent share of 0.9 percent, and an access-barrier rate of 10.1 percent.

Table 4. MEPS 2022 medical out-of-pocket burden by poverty/income category. Person-level weighted estimates from HC-243. OOP is T0TSLF22 ; income is FAMINC22 ; access barrier equals any delayed/unaffordable medical, dental, or prescription care indicator observed in round 4/2.

Group	N	Weighted pop.	Mean family income	Mean OOP medical	Mean OOP / mean income %	Median OOP/income %	P90 OOP/income %	OOP $\geq 10\%$ income %	Any access barrier %
Poor/negative income	3,626	38.3M	\$11,029	\$375	3.40	0.03	11.26	10.9	20.4
Near poor	1,017	12.5M	\$26,978	\$681	2.52	0.06	6.80	7.3	20.5
Low income	3,000	41.2M	\$40,213	\$697	1.73	0.10	5.84	5.6	18.7
Middle income	6,054	98.7M	\$71,885	\$814	1.13	0.18	3.67	2.4	16.0
High income	8,050	142.3M	\$179,328	\$1,228	0.68	0.20	2.03	0.9	10.1

Access barriers also rise across the OOP-burden distribution. In MEPS 2022, the highest OOP-burden quartile had a 24.0 percent access-barrier rate, compared with 8.9 percent in the lowest quartile. This pattern is descriptive, not causal. Higher OOP burden can reflect worse health, poorer coverage, local access constraints, or selection into plan types. The table shows why a resilience metric cannot stop at average spending; it does not show that OOP burden itself causes access barriers.

Table 5. MEPS 2022 access barriers by out-of-pocket medical burden quartile. Positive-family-income persons only. Descriptive, not causal; higher burden can reflect worse health as well as prices or coverage.

OOP burden quartile	N	Weighted pop.	Mean OOP medical	Median OOP/income %	Mean OOP/income %	Any access barrier %	Fair/poor health %
Q1	5,665	91.9M	\$0	0.00	0.00	8.9	5.5
Q2	4,057	70.4M	\$92	0.05	0.06	9.6	7.2
Q3	5,145	81.0M	\$548	0.43	0.47	14.9	9.8
Q4	6,070	81.1M	\$3,112	2.55	31.89	24.0	16.9

The multi-year MEPS robustness table reinforces the point. From 2019 through 2023, any access barrier ranged from 13.0 to 18.7 percent, while the joint profile combining high OOP burden, access barriers, and uninsured status ranged from 19.9 to 24.8 percent. The relevant household-risk object is therefore broader than spending observed in a single category.

Table 6. MEPS 2019–2023 medical-fragility robustness benchmarks. Person-weighted estimates from Full-Year Consolidated public-use files. Bracketed entries use an approximate Taylor-style $\text{VARSTR} / \text{VARPSU}$ linearization. The joint profile flags OOP ≥ 10 percent of family income, any delayed/unaffordable medical/dental/prescription care, or uninsured status.

Year	N	Weighted pop.	Any access barrier %	OOP $\geq 10\%$ %	Zero OOP + barrier %	Joint profile %	Uninsured %	Uninsured barrier % [CI]	Barrier among zero-OOP % [CI]	P90 burden %
2019	28,512	323.1M	18.7	3.1	3.6	24.8	6.3	40.5 [37.1, 43.9]	13.2 [12.1, 14.3]	3.37
2020	27,805	324.8M	15.1	3.0	2.9	21.7	6.5	31.5 [28.1, 34.8]	10.0 [9.0, 11.1]	3.17
2021	28,336	327.2M	13.0	3.2	2.2	19.9	6.1	26.5 [23.0, 30.0]	7.8 [7.0, 8.6]	3.46
2022	22,431	328.9M	14.5	3.1	2.6	21.4	6.4	30.0 [26.0, 34.0]	9.0 [7.7, 10.2]	3.30
2023	18,919	330.0M	15.6	2.8	3.1	22.4	6.3	28.7 [25.0, 32.5]	11.3 [10.1, 12.6]	3.22

VII. Interpretation: the hidden fragility premium

The revised evidence supports a more disciplined version of the hidden-fragility-premium idea. The term should not be used as a slogan for any fast-rising insurance price. It should describe the recurring cost of maintaining protection against shocks, scaled by the household's capacity to absorb those costs.

Three findings matter.

First, selected defensive categories have grown faster than headline CPI in aggregate data, especially auto insurance and hospital services. This explains why households may continue to feel pressure after headline inflation cools.

Second, the CEX burden exercise shows that selected resilience costs are distributional. Lower-income consumer units spend less in dollars but more as a share of income. That is the central household-finance fact.

Third, MEPS shows that medical-risk exposure cannot be measured with spending alone. Low realized out-of-pocket spending may indicate good coverage or low need, but it may also reflect delayed or unaffordable care. A serious fragility premium must include both realized cost and unmet-need indicators.

These findings matter for household planning. A household trying to preserve long-duration savings cannot evaluate durability from income and headline inflation alone. It also needs to know whether recurring protection costs, deductibles, medical exposure, and vehicle or housing coverage are crowding out liquid buffers.

VIII. Outcome validation is the next empirical test

The burden concept becomes economically important only if it predicts household outcomes. The current note does not yet show that insurance-resilience burden reduces emergency savings, increases revolving credit, raises delinquency risk, causes missed payments, leads households to reduce coverage, or forces asset sales. Those are the outcomes that would turn the hidden fragility premium from a useful measurement frame into a stronger household-finance contribution.

The most realistic next validation step is to triangulate across public household datasets. CEX can describe selected spending burden by consumer unit. MEPS can describe medical-risk burden and access barriers. SHED can describe emergency-expense capacity, cash-equivalent payment ability, bill pressure, and hardship. SCF can describe liquid assets and balance-sheet heterogeneity. A stronger paper would either link these directly where possible or build a transparent validation bridge: higher insurance/medical-risk burden should be associated with weaker emergency-buffer capacity, more borrowing reliance, more skipped care, or higher missed-payment risk.

IX. What the paper still does not prove

The paper remains short of a causal journal article. It does not show that insurance repricing causes lower saving, higher credit-card debt, missed payments, reduced coverage, or asset liquidation. It does not exploit state-level premium shocks, disaster exposure, employer plan variation, or the 2022–2024 auto-insurance repricing episode as an identification design. It does not directly observe liquid resources in the CEX burden tables used here.

Those are not minor gaps. They define the next version of the research agenda. A stronger paper would link state-level auto or property insurance shocks to household credit, delinquency, coverage, or emergency-saving outcomes. Another path would merge or triangulate CEX, SHED, SCF, and MEPS evidence to estimate how many households face insurance-resilience burden above a liquidity threshold.

The present draft should therefore be understood as a public-data measurement paper. It turns the concept into an estimable burden object, shows distributional burden using CEX, and connects medical-risk burden to access barriers using MEPS. That is a stronger contribution than the earlier version, but it is still descriptive.

X. Conclusion

Insurance and medical-risk costs are part of household resilience, not merely line items in a budget. Aggregate price indexes show why the issue is timely, but the more important evidence is distributional: selected resilience costs absorb a larger income share for lower-income consumer units, and medical-risk burden is linked to access barriers that spending alone can miss.

The appropriate conclusion is narrow. Public data support measuring an insurance-resilience burden as part of household fragility analysis. They do not yet prove a causal insurance-shock channel. The next research step is to connect that burden to liquidity, borrowing, coverage choices, and distress outcomes.

For practical household resilience tools, the implication is immediate: insurance and medical-risk exposure should be asked about directly. A household's ability to sustain saving depends not only on income and headline inflation, but on whether recurring protection costs leave enough liquid capacity to withstand the next shock.

Appendix A. Methods notes for the current measurement draft

A.1 Aggregate BLS/FRED module

Monthly CPI series are converted to calendar-year annual averages. Growth is calculated as $(\text{end annual average} / \text{start annual average} - 1) * 100$. CAGR is calculated as $(\text{end} / \text{start})^{(1 / \text{years})} - 1$. CPI rows are price indexes. Consumer Expenditure rows in the aggregate motivation table are nominal annual spending series for all consumer units and are not price indexes. Seasonal-adjustment status differs across series; annual averaging reduces but does not eliminate this concern.

A.2 CEX burden module

The CEX module uses calendar-year 2024 public-use Consumer Expenditure work developed for the broader fragility-premium project. The unit is the consumer unit. Income groups are based on reported annual income. The core numerator in this note combines selected reference-period health-insurance, healthcare out-of-pocket, vehicle-insurance, and housing-insurance costs. Childcare is excluded from the core burden measure. Burden ratios divide selected reference-period costs by annual income or total expenditure, so the estimates are descriptive burden screens rather than fully annualized household-cost estimates. The current note reports point estimates only. A full research paper should add survey-design standard errors, replicate-weight uncertainty where available, exposed-unit estimates, and robustness to positive-income restrictions and missing/negative-income treatment.

A.3 MEPS medical-risk module

The MEPS module uses public-use estimates from prior medical-fragility work. OOP burden tables use medical out-of-pocket spending divided by family income. Poor or negative-income groups create denominator sensitivity, and high-burden means are sensitive to right-tail observations. A full paper should report winsorized estimates, positive-income restrictions, family-level aggregation, survey-design standard errors, and alternative burden thresholds. Access-barrier measures are descriptive indicators of delayed or unaffordable medical, dental, or prescription care; they should not be interpreted as causal effects of out-of-pocket burden.

References

- Board of Governors of the Federal Reserve System. 2025. *Economic Well-Being of U.S. Households in 2024*. Washington, DC: Board of Governors.
- Federal Reserve Bank of St. Louis. 2026. *FRED Economic Data*. St. Louis, MO.
- Kaplan, Greg, Giovanni L. Violante, and Justin Weidner. 2014. "The Wealthy Hand-to-Mouth." *Brookings Papers on Economic Activity* 2014 (1): 77–138.
- Lusardi, Annamaria, Daniel Schneider, and Peter Tufano. 2011. "Financially Fragile Households: Evidence and Implications." *Brookings Papers on Economic Activity* 2011 (1): 83–134.
- U.S. Bureau of Labor Statistics. 2025. *Consumer Expenditure Surveys*. Washington, DC: U.S. Bureau of Labor Statistics.
- U.S. Bureau of Labor Statistics. 2026. *Consumer Price Index*. Washington, DC: U.S. Bureau of Labor Statistics.
- U.S. Department of Health and Human Services, Agency for Healthcare Research and Quality. 2024. *Medical Expenditure Panel Survey Household Component Public Use Files*. Rockville, MD.